PROJECT TITLE :

Automated Log File Analyzer for Intrusion Detection and Threat Pattern Analysis Using Python.

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PROGRAM: Cybersecurity and Digital Forensic Engineering

YEAR: 2026

Abstract

This project develops a Python-based log file analyzer for intrusion detection. It processes SSH, Apache, Nginx, Syslog and firewall logs to identify suspicious IP behavior, detect brute-force attacks and generate structured security reports.

Abstract - - -

CHAPTER 1: INTRODUCTION

1.1 Background

Modern systems generate large volumes of log data that are difficult to analyze manually. Cyber attackers often exploit this by hiding malicious activity in logs.

1.2 Problem Statement

Manual log analysis is inefficient, time-consuming and prone to human error.

1.3 Significance

This system automates intrusion detection and supports cybersecurity analysts.

CHAPTER 2: OBJECTIVES

2.1 Main Objective

To develop an automated log analysis system for intrusion detection.

2.2 Specific Objectives

- Detect log types automatically
- Extract IP addresses
- Identify suspicious behavior
- Generate forensic reports

CHAPTER 3: LITERATURE REVIEW

Log analysis tools like Splunk and ELK Stack are widely used in SOC environments. However, they are complex and resource-intensive. This project provides a lightweight alternative for educational and forensic purposes.

CHAPTER 4: METHODOLOGY

4.1 Approach

A rule-based detection system using Python.

4.2 Tools Used

- Python 3
- Regex
- Linux terminal
- Sample log files

4.3 Process Flow

1. Load log file
2. Detect log type
3. Scan for keywords
4. Extract IP addresses
5. Count occurrences
6. Generate report

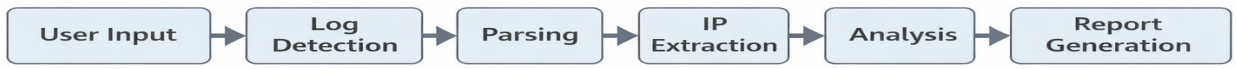

Figure 1: System Workflow

CHAPTER 5: SYSTEM DESIGN

5.1 Architecture

The system follows a modular pipeline design.

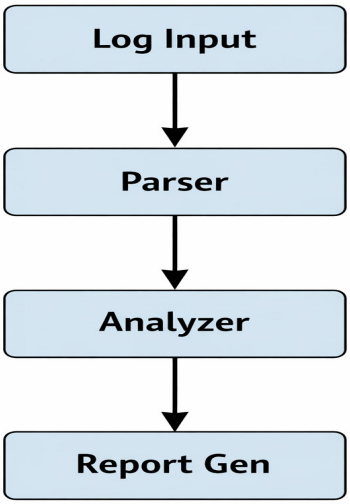

Figure 2: Architecture Diagram

CHAPTER 6: IMPLEMENTATION

6.1 Language

Python 3

6.2 Core Modules

- Log detection module
- IP extraction module
- Threat detection engine
- Report generator

```
┌──(alfred㉿xscape404)-[~/CyberSecurity/03_PROJECTS/Personal_Projects/log-analyzer-project]
└─$ python3 analyzer.py

═══════════════════════════════════════════════════════════
 LOG FILE ANALYZER v2.1
 SOC / Cybersecurity Monitoring Tool
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Enter log file path: sample_logs/ssh.log
[!] Using generic log profile
[✓] Loaded 7143 lines from 'sample_logs/ssh.log'
[✓] Suspicious lines: 5069
[✓] Success events: 3619
[⚠] Suspicious IPs detected: 29

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 LOG FILE ANALYZER - SECURITY REPORT
 File     : ssh.log
 Log Type : Generic / Unknown Log
 Generated: 2026-04-17 07:57:23
═══════════════════════════════════════════════════════════

[1] OVERVIEW
 Total lines          : 7143
 Suspicious lines     : 5069
 Unique IPs           : 36
 Suspicious IPs       : 29
 Compromised IPs      : 28

[2] TOP OFFENDING IPS
 192.168.202.141    2365 events ⚠
 192.168.202.110    613 events ⚠
 192.168.204.45     574 events ⚠
 192.168.202.140    433 events ⚠
 192.168.202.68     161 events ⚠
 192.168.202.138    142 events ⚠
 192.168.203.45     129 events ⚠
 192.168.202.109    113 events ⚠
```

Figure 3: Execution of log analyzer detecting SSH brute-force attempts

CHAPTER 7: TESTING AND RESULTS

7.1 Test Data

Sample SSH logs containing brute-force attempts.

7.2 Output Results

- Suspicious IPs detected
- Compromised IPs identified
- Report generated successfully

```
┌──(alfred㉿xscape404)-[~/CyberSecurity/03_PROJECTS/Personal_Projects/log-analyzer-project]
└─$ cat security_report.txt

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 LOG FILE ANALYZER - SECURITY REPORT
 File     : ssh.log
 Log Type : Generic / Unknown Log
 Generated: 2026-04-17 08:03:15
═══════════════════════════════════════════════════════

[1] OVERVIEW
 Total lines            : 7143
 Suspicious lines       : 5069
 Unique IPs             : 36
 Suspicious IPs         : 29
 Compromised IPs        : 28

[2] TOP OFFENDING IPS
 192.168.202.141    2365 events ⚠
 192.168.202.110    613 events ⚠
 192.168.204.45     574 events ⚠
 192.168.202.140    433 events ⚠
 192.168.202.68     161 events ⚠
 192.168.202.138    142 events ⚠
 192.168.203.45     129 events ⚠
 192.168.202.109    113 events ⚠
 192.168.202.79     90 events ⚠
 192.168.202.112    90 events ⚠

[3] SUSPICIOUS IPS (≥ 3)
 ⚠ 192.168.202.68 → 161 events
 ⚠ 192.168.202.79 → 90 events
 ⚠ 192.168.202.109 → 113 events
 ⚠ 192.168.204.45 → 574 events
 ⚠ 192.168.203.45 → 129 events
```

Figure 4: Generated security report showing suspicious IP analysis

```
[4] COMPROMISED IPS
ALERT: 192.168.202.[illegible] had both failure + success
ALERT: 192.168.202.79 had both failure + success
ALERT: 192.168.202.109 had both failure + success
ALERT: 192.168.204.45 had both failure + success
ALERT: 192.168.203.45 had both failure + success
ALERT: 192.168.202.110 had both failure + success
ALERT: 192.168.202.76 had both failure + success
ALERT: 192.168.203.61 had both failure + success
ALERT: 192.168.202.103 had both failure + success
ALERT: 192.168.202.87 had both failure + success
ALERT: 192.168.202.100 had both failure + success
ALERT: 192.168.202.112 had both failure + success
ALERT: 192.168.202.108 had both failure + success
ALERT: 192.168.202.81 had both failure + success
ALERT: 192.168.202.80 had both failure + success
ALERT: 192.168.202.95 had both failure + success
ALERT: 192.168.202.119 had both failure + success
ALERT: 192.168.202.90 had both failure + success
ALERT: 192.168.202.125 had both failure + success
ALERT: 192.168.202.4 had both failure + success
ALERT: 192.168.202.138 had both failure + success
ALERT: 192.168.202.140 had both failure + success
ALERT: 192.168.202.98 had both failure + success
ALERT: 192.168.202.144 had both failure + success
ALERT: 192.168.202.136 had both failure + success
ALERT: 192.168.202.69 had both failure + success
ALERT: 192.168.202.150 had both failure + success
ALERT: 192.168.202.141 had both failure + success
```

Figure 5: SSH log file containing failed login attempts

CHAPTER 8: DISCUSSION

The system successfully identifies brute-force patterns using threshold-based detection. However, it may generate false positives in noisy environments.

CHAPTER 9: LIMITATIONS

- No machine learning model
- No real-time streaming
- No external threat intelligence integration

CHAPTER 10: CONCLUSION

The project successfully demonstrates a functional intrusion detection prototype suitable for cybersecurity education and SOC simulation environments.

CHAPTER 11: RECOMMENDATIONS

Future improvements include:

- Integration with SIEM systems
- Machine learning anomaly detection
- Real-time monitoring
- Web dashboard interface

CHAPTER 12:REFERENCES

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The Practice of Network Security Monitoring. (2013). No Starch Press.

Python for Cybersecurity. (2019). Packt Publishing.